Prathyush Sajith

psaji@uic.edu | prathyushsajith17@gmail.com | Linkedin | Portfolio
Chicago, Illinois - 60612

Education

- **University of Illinois at Chicago** *Aug 2026 (expected)*
  MS Thesis in Computer Engineering
  - Advisor: Prof. Ahmet Enis Cetin
  - Selected Courses: Convex Optimization, Neural Networks, Image Analysis and CV, Pattern Recognition.
- **SRM Institute of Science and Technology, Ramapuram** *May 2024*
  B.Tech in Computer Science Engineering
  - Grade: 9.20/10.00 (First class with Distinction)

Experience

Graduate Research Assistant – Machine Learning & Medical Imaging *Sep 2025 – Present*
[Tools used: Python - Pandas, PyTorch, Scikit-learn, OpenCV, quPATH, ImageJ // HIPAA certified]

- Developed an end-to-end workflow integrating QuPath-based multi-channel fluorescence image quantification with statistical pipelines to extract, normalize, and compare cell population across experimental conditions.
- Fine-tuned and deployed a Mask R-CNN instance segmentation model via Amazon SageMaker JumpStart to automate detection and morphological analysis of cellular structures in high-resolution but noisy TIRFM imagery.
- Used Pretrained models to reconstruct high-resolution 3D mappings of mule brain structures from TIRFM images.

MS Thesis

Improving Efficiency of Stereo Depth Estimation using Walsh-Hadamard Transforms *May 2025 – Present*
[Tools used: PyTorch, TorchVision, OpenCV, CUDA, cuDNN, PIL, Matplotlib]

- Designed and extended a **Stereo Transformer (STTR)** based stereo depth estimation pipeline on a custom **CARLA** synthetic dataset, leveraging attention-driven correspondence modeling along epipolar lines for disparity estimation.
- Improved computational efficiency by replacing standard Conv2D layers with a Walsh-Hadamard Transform-based convolution (WHTConv2D) **by 16%**, reducing complexity while preserving representational capacity.
- Implemented a **Log-Disparity Loss formulation** to emphasize supervision on far-range (>100 m) depth regions, improving stability and accuracy for small-disparity predictions.

Selected Projects

(1) History-Aware Local RAG System with LangChain & Ollama (git) *Apr 2026*
[Tools used: Python, LangChain (LCEL), Ollama (Llama 3.2), HuggingFace Embeddings (all-MiniLM-L6-v2), FAISS]
Built a fully local Retrieval-Augmented Generation system to query about 3.4M characters of literary text with no external API dependencies. Chunked documents into 4,700+ segments, embedded them using HuggingFace's all-MiniLM-L6-v2, and indexed with FAISS for semantic search. Implemented MMR-based retrieval and a conversational interface with history-aware question rewriting using LangChain LCEL and a locally hosted Llama 3.2 via Ollama.

(2) Custom built GPT-Based Large Language Model (git) *Jan – Feb 2025*
[Tools used: Pytorch, Hugging Face Transformers, NumPy, Pandas]
Built a 124M parameter GPT-based LLM using Pytorch. Designed an efficient tokenizer, created embeddings with 768-1280 dimensions, and implemented multi-head self-attention and rotary embeddings. Pretrained the model on up to 1024 tokens per context window and achieved significant loss reduction. Fine-tuned the model to generate coherent text outputs, by data preprocessing and optimization.

(3) Lane Detection using UNet implemented as Discrete Wavelet Transform (git) *Nov – Dec 2024*
[Tools used: Python, Keras, TensorFlow-GPU, OpenCV // Architecture: Discrete Wavelet Transform, U-Net]
Developed a deep learning model for lane detection that integrates UNet architecture as a Discrete Wavelet Transform (DWT). It enhances detection accuracy, while reducing the aliasing effects, that's usually found during the upsampling process. It is implemented using GPU kernel programming in CUDA.

Publications

(1) Crowd analysis and tracking for individual rescue operation using machine learning. IJRASET 2024 | ResearchGate

Skills

- **Languages:** Python, SQL, MATLAB, (familiar with C and C++).
- **Deep Learning:** Tensorflow & Keras, Pytorch, Scikit-learn, Numpy, LangChain, RAG, Pandas, OpenCV.
- **Analysis:** Matlab, Tableau, Microsoft Power BI, Matplotlib and Seaborn, Excel, Qualtrics, Forms, Data Cleaning, EDA.
- **Containers and Orchestration:** Docker, Kubernetes.